

30	A photon of wavelength 555.00 nm, measured to an accuracy of 0.01 nm. What is the minimum uncertainty in the location of the photon?					
	A	$6.6 \times 10^{-23} \text{ m}$	B	$1.0 \times 10^{-11} \text{ m}$	C	$3.1 \times 10^{-2} \text{ m}$
					D	$3.0 \times 10^{-1} \text{ m}$
	Answer: C De Broglie Wavelength, $p = \frac{h}{\lambda}$ $\frac{\Delta p}{p} = \frac{\Delta h}{h} + \frac{\Delta \lambda}{\lambda} = \frac{\Delta \lambda}{\lambda}$ $\Delta p = \frac{\Delta \lambda}{\lambda} p = \frac{0.01}{555.00} \left(\frac{6.63 \times 10^{-34}}{555.00 \times 10^{-9}} \right) = 2.1524 \times 10^{-32} \text{ N m}$ Heisenberg Uncertainty Principle, $\Delta p \Delta x \gtrsim h$ $2.1524 \times 10^{-32} \Delta x \gtrsim 6.63 \times 10^{-34}$ $\Delta x \gtrsim 3.1 \times 10^{-2} \text{ m}$					

